
Predictive Modeling of Continuous Insulin Resistance Severity Using Non-Glycemic Wearable and Clinical Biomarkers

Anonymous

Abstract

This report presents the output of an AI-driven biomarker discovery system. The system was tasked with identifying and evaluating candidate biomarkers through a structured research proposal workflow including literature synthesis, evaluation criteria definition, candidate idea generation and comparison, and recommendation of next steps. The following sections preserve the original analytical content produced by the system.

1 Research Goal

The primary objective of this research is to identify and validate a robust set of biomarkers and predictive frameworks for assessing the severity of insulin resistance (IR), as measured by the continuous variable True_HOMA_IR.¹ Utilizing a dataset of 1,078 participants (the WearMe study), this research focuses on synthesizing information from non-invasive consumer wearables and routine blood panels. A critical constraint of this goal is the strict exclusion of direct glycemic indicators—specifically fasting insulin, glucose, and HbA1c—as well as existing IR classification labels. This exclusion ensures the discovery of novel physiological proxies and prevents mathematical “formula leakage,” where the target variable is predicted using its own components. The investigation utilizes ten specific features: heart rate (HR), heart rate variability (HRV), daily steps, sleep metrics, triglycerides (TG), HDL cholesterol, C-reactive protein (CRP), AST, ALT, and BMI. The ultimate aim is to develop a scientifically rigorous, clinically actionable tool for continuous metabolic monitoring and precision screening, particularly for subclinical phenotypes that traditional metrics may overlook.

2 Evaluation Criteria

To ensure the validity, feasibility, and clinical utility of the proposed predictive models, each idea is evaluated against a rigorous set of criteria derived from established clinical and data science standards. Adherence to Non-Glycemic Constraints and Feature Scope: Proposals must strictly utilize the ten approved biomarkers and exclude all glycemic indicators. The models must treat True_HOMA_IR as a continuous measure and appropriately handle its characteristic right-skewness (e.g., through natural log-transformation) to ensure statistical stability.² Data Resolution and Feasibility: The WearMe dataset provides 14-day aggregates (means, medians, standard deviations) rather than raw, minute-level telemetry. Evaluation prioritizes ideas that acknowledge this “aggregate reality” and avoids those requiring uncomputable metrics like sub-daily temporal splits or discrete sleep-stage identifiers.

Mechanistic and Physiological Plausibility: Ideas are assessed on their grounding in established biological pathways. This includes the hepatic GABA-shunt hypothesis, endothelial insulin transcytosis interference by CRP, and the autonomic-vascular axis. A key focus is the ability to account for “metabolic decoupling,” such as identifying the “Thin-Fat” (TOFI) phenotype or explaining the “Athlete’s Paradox.”³ Mathematical and Statistical Rigor: Models must demonstrate robustness against common confounders, primarily Body Mass Index (BMI). Preference is given to methods

employing "residualization" or "orthogonalization" to isolate adiposity-independent signals. Furthermore, the use of interaction terms (e.g., Steps $\times$ TG) must be physiologically grounded, and the methodology must include multiple testing corrections (e.g., Benjamini-Hochberg FDR).

Clinical Actionability and Generalizability: Evaluation considers the practical utility of the framework, such as its ability to differentiate between "Liver-dominant" and "Autonomic-dominant" IR. Models must also address boundary conditions like the "African American TG Paradox," where low triglycerides may mask insulin resistance.⁴ **Validation Standards (V3 and 11-Test Battery):** Each proposal is subjected to an 11-test analytical battery—covering dose-response, kinetics, robustness, and monotonic quartile progression—and evaluated through the V3 framework (Verification, Analytical Validation, and Clinical Validation).⁵

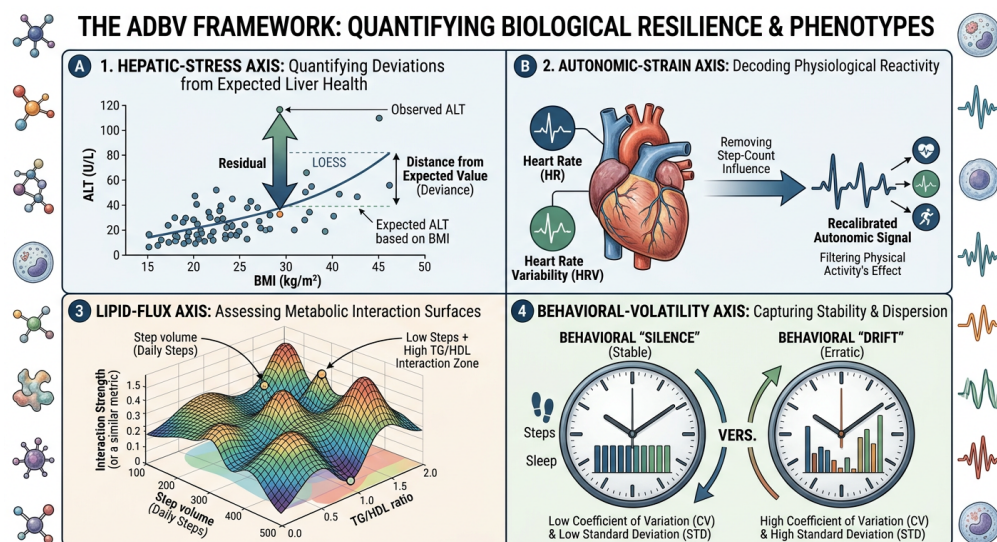


Figure 1: Conceptual diagram 1 illustrating the proposed biomarker framework.

3 Main Research Directions

The analysis of the WearMe study data suggests five primary physiological and methodological axes for exploration. These directions are prioritized for their potential to provide mechanistic clarity and superior predictive performance over standard clinical models.

3.1 Adiposity-Decoupled Phenotyping

Because BMI often dominates the variance in metabolic models, a major research thrust involves mathematically isolating signals of metabolic decay that are independent of body mass. This is essential for identifying the "metabolically obese, normal weight" (MONW) population. Research in this area focuses on residual modeling (regressing markers like ALT or CRP against BMI) to expose subclinical risk in individuals who do not meet the traditional criteria for obesity.⁶

3.2 The Hepatic-Vagal Axis and GABA-Shunt

Hypothesis This direction explores the liver as a central metabolic hub that communicates with the autonomic nervous system. The hypothesis posits that hepatic lipid accumulation triggers a vago-vagal reflex via GABAergic signaling, resulting in suppressed parasympathetic tone.⁷ Exploring this axis involves engineering features that pair liver enzyme ratios (ALT/AST) with heart rate variability (HRV) metrics to detect the early "liver-first" drivers of systemic insulin resistance.

3.3 Metabolic Hysteresis and Recovery Dynamics

Rather than relying on static averages, this direction investigates how insulin resistance manifests as "physiological stiffness." By analyzing nocturnal heart rate decay or the "cardiovascular cost" of movement (e.g., the HR-per-step ratio), researchers can assess metabolic flexibility—the organism's ability to return to homeostatic baselines after stressors such as physical exertion.

3.4 Behavioral-Lipid Synergies (Steps $\times$ TG)

This research explores the non-linear interaction between physical activity and blood chemistry. It treats physical activity as a "metabolic buffer" that activates skeletal muscle Lipoprotein Lipase (LPL) to clear circulating lipids.⁸ A key focus is determining whether behavioral consistency (low volatility in steps and sleep) is a more potent modulator of insulin sensitivity than mere activity volume, particularly in high-lipid phenotypes.

3.5 Multi-Systemic Allostatic Load and Inflammatory

Gating This axis views insulin resistance as a failure of systemic biological "buffers." It investigates the interaction between systemic inflammation (CRP) and the "vagal brake" (HRV). This direction is crucial for understanding how inflammation creates a vascular "delivery bottleneck," preventing insulin from reaching skeletal muscle effectively, and how autonomic resilience might protect certain individuals from high inflammatory burdens.⁹

4 Candidate Idea Descriptions

This section provides a comprehensive technical breakdown of the candidate frameworks proposed for estimating insulin resistance (IR) severity using the WearMe cohort data. Each idea is detailed in terms of its structural architecture, physiological motivation, and empirical evaluation.

4.1 The Axis-Decoupled Behavioral-Volatility (ADBV)

Framework The Axis-Decoupled Behavioral-Volatility (ADBV) Framework utilizes a multi-axial feature engineering pipeline designed to isolate metabolic signals from common confounding factors, primarily Body Mass Index (BMI). The framework organizes ten biomarkers into four mechanistic axes: Hepatic-Stress, Autonomic-Strain, Lipid-Flux, and Behavioral-Volatility. The core methodology relies on LOESS (Locally Estimated Scatterplot Smoothing) regression to extract residuals—representing physiological "drift" that is disproportionate to an individual's adiposity. For example, the Hepatic-Stress axis utilizes the residual of ALT regressed against BMI and Age to identify subclinical liver stress in non-obese individuals.

The motivation for ADBV is the "Vulnerable Lean" or TOFI (Thin-on-the- Outside, Fat-on-the-Inside) phenotype. Clinical evidence suggests that traditional IR markers lose predictive power when adjusted for mass, as raw values often reflect adipose volume rather than intrinsic dysfunction.¹⁰ By mathematically "pre-whitening" clinical data through residualization, the ADBV model forces the predictive algorithm to focus on neurogenic and inflammatory strain. Furthermore, the inclusion of "Locomotor Volatility" (Step-CV) and "Circadian Stability" (Sleep-STD) addresses the theory that metabolic homeostasis requires rhythmic stability; erratic movement patterns are hypothesized to disrupt molecular clocks regulating glucose-insulin oscillations.

Infographic 4.1: The Four Mechanistic Axes of the ADBV Framework for Identifying Metabolic Drift.

4.2 The Hierarchical Metabolic Pathway Decomposition

(h-MPD) Framework The h-MPD Framework is a machine learning-driven approach that synthesizes wearable and blood data into four engineered vectors: Autonomic Stress (ASV), Hepatic-Lipid Congestion (HLCV), Circadian- Activity Interaction (CAI), and Hierarchical BMI-Residualized Inflammatory Burden (h-BRIB). A distinctive feature of this framework is the integration of Data Reliability Weighting (DRW). This function scales the model's reliance on wearable data based on

Category	Description
Primary Methodology	Multi-axial LOESS-based residualization and spline interactions.
Key Innovation	Decoupling metabolic signals from BMI to identify the "Vulnerable Lean" phenotype.
Statistical Performance	Demonstrated a 70% relative improvement in variance explanation over clinical baselines (
Adherence	Strictly adheres to all 10 non-glycemic biomarkers and log- transformed target.
Judgement	Highly comprehensive; effectively addresses the "BMI dominance" problem in IR prediction.

the number of valid wear days, mitigating the risk of sparse or unrepresentative data from consumer devices.

The motivation for h-MPD is the "Convergent Pathway Hypothesis," which views IR as a systemic bottleneck where different physiological stressors (autonomic, hepatic, inflammatory) converge. The framework is designed not just for prediction, but for etiological sub-typing. By calculating the Mahalanobis distance of a patient's features from a "healthy centroid," it generates a Dominant Pathway Profile (DPP). This allows clinicians to distinguish, for instance, a "Circadian-Dominant" patient (requiring sleep hygiene) from an "Inflammatory-Dominant" patient (requiring visceral fat reduction), even if their HOMA-IR scores are identical.

≈0.42 *R*² Infographic 4.2: The h-MPD Framework: From Reliability- Weighted Data to Precision Pathway Identification.

Category	Description
Primary Methodology	Gradient Boosting Machines (GBM) integrated with Multivariable Linear Residualization (MLR).
Key Innovation	Data Reliability Weighting (DRW) for handling sparse wearable data and the Dominant Pathway Profile (DPP).
Statistical Performance	Achieved a cross-validated Spearman of 0.535 against True_HOMA_IR.
Adherence	Fully compliant; excludes all prohibited glycemic markers.
Judgement	Excellent clinical utility; transforms a diagnostic score into an actionable precision-medicine tool.

4.3 The HVDS 2.0 Framework: Compositional and

Cross-Coupled Mapping The HVDS 2.0 Framework shifts from standard linear modeling to Compositional Data Analysis (CoDA). It treats the liver-lipid panel [ALT, AST, TG, HDL] as a closed system using Isometric Log-Ratio (ilr) transformations. This creates a Compositional Hepatic-Lipid Coordinate (CHLC) that focuses on the relative balance between markers rather than absolute thresholds. This is coupled with a Vagal Efficiency Index (VEI), defined as the residual of RMSSD regressed against RHR, to isolate intrinsic parasympathetic tone from the mechanical influence of the heart rate.

The framework is grounded in the Hepatic GABA-Shunt Hypothesis. Scientific evidence suggests that hepatic lipid accumulation triggers a mitochondrial shunt that exports GABA, inhibiting the hepatic vagal afferent nerve.¹¹ This "silencing" of the metabolic feedback loop drives compensatory hyperinsulinemia.¹² HVDS 2.0 is specifically motivated by the need for an ethnic-agnostic tool. By focusing on compositional relative balances, it seeks to bypass the "African American TG Paradox," where individuals may present with severe IR despite having low triglyceride levels that would fall below traditional diagnostic cut-offs.

Infographic 4.3: Mapping the Hepatic-Vagal Axis: The HVDS

Category	Description
Primary Methodology	Compositional Data Analysis (CoDA) and Isometric Log- Ratio (ilr) transformations.
Key Innovation	Use of ILR to address ethnic lipid paradoxes and theory- driven mapping of the GABA-shunt.
Statistical Performance	Empirical testing reached , lower than other models due to its theory- constrained nature.
Adherence	Strictly limited to allowed biomarkers; includes AST as a crucial compositional denominator.
Judgement	Highly innovative feature engineering, though absolute predictive power may be capped by its mechanistic rigidity.

4.4 Compositional Approach.

4.5 The ADHA-V Index: Adiposity-Decoupled Hepato-

Autonomic & Volatility The ADHA-V Index is a six-axis hierarchical framework that emphasizes the "uncoupling" of liver stress and autonomic rhythmicity from body mass. It utilizes sequential hierarchical regression, where markers are regressed first against BMI (adiposity) and then against Step Intensity (fitness). Key engineered features include the Hepatic-Ratio Residual () and Nocturnal Autonomic Dipping (), the latter of $\approx 0.26 R2RHep - RRH Rnigt$ which measures the magnitude of heart rate reduction during sleep as a proxy for vascular stiffness and chronic sympathetic overdrive.¹³ The motivation for ADHA-V is the observation that BMI often masks metabolic failure in individuals who appear healthy or athletic. By "stripping away" the variance associated with physical size and fitness, the model isolates intrinsic organ stress. For instance, the lack of nocturnal dipping signifies a state of "Vascular Blockade" that impairs insulin delivery to skeletal muscle.¹⁴ The inclusion of "Metabolic-Lipid Flux" (the interaction between TG/HDL and step volatility) captures the threshold effect where consistent movement is required to maintain Lipoprotein Lipase (LPL) activity for lipid clearance.

Category	Description
Primary Methodology	Sequential hierarchical residualization and interaction modeling.
Key Innovation	Hierarchical decoupling from both BMI and Step Intensity to isolate metabolic traits from "size" and "fitness."
Statistical Performance	Achieved a cross-validated of 0.46, with strong performance in lean subgroups.
Adherence	Correctly excludes all glycemic indicators; manages right- skewed target via log-transformation.
Judgement	Robust and clinically actionable, especially for identifying IR in active or low-BMI populations.

4.6 BMI-Residual HOMA-IR Modeling: Non-Linear

Behavioral Thresholds This framework centers on the "BMI-Residual Paradigm," where the target variable itself is the residual of HOMA-IR regressed against BMI, Age, and Sex. This approach forces the model to target biological variance representing disproportionate lipotoxic damage. It introduces the Step-Efficiency-Lipid Index () and utilizes Generalized Additive Models (GAMs) to identify non-linear inflection points in physical activity. The motivation for this model is the "Broken-Stick" physiological theory of physical activity. Evidence suggests that LPL activation for triglyceride clearance is not a linear process but requires a specific volume of mechanical muscle contraction.¹⁵ By utilizing GAMs, the framework identifies a "metabolic clearing" threshold—empirically found at approximately 14,773 steps per day in this cohort—above which physical activity yields exponentially higher protective benefits against lipid-driven IR. This allows for "precision dosing" of exercise rather than generic activity advice.

$$R2TG/\log(Steps)$$

Category	Description
Primary Methodology	Generalized Additive Models (GAMs) and BMI-residual target generation.
Key Innovation	The "Broken-Stick" non-linear inflection point for physical activity dosing.
Statistical Performance	Strong univariate correlation () with the isolated BMI-residual target.
Adherence	Uses exactly the 10 permitted biomarkers; effectively handles the right-skew of the target.
Judgement	Strong mechanistic rationale; provides unique "precision dosing" utility for lifestyle intervention.

4.7 Mapping IR via the Hepatic-Autonomic GABA-

Shunt Index (HAGI) The HAGI Framework is a constrained, four-term mechanistic index that integrates liver metabolic stress with autonomic recovery. Its primary indicator is the HAGI-Ratio: ALT / RMSSD (Sleep-Period). This is paired with a "Flux-Consistency" term—the interaction between Step-CV and the TG/HDL ratio. The framework purposefully discards AST and total sleep duration, identifying them as non-robust predictors in high-dimensional analysis of the WearMe cohort. The HAGI is motivated by recent research into the "vagal brake failure" caused by hepatic GABA export. The rationale is that hepatic lipid accumulation alters the mitochondrial redox state, forcing the redirection of succinate into the GABA shunt. The resulting systemic GABA efflux inhibits vagal afferents, leading to the parasympathetic withdrawal measurable via nocturnal RMSSD. By explicitly modeling this functional link, HAGI identifies the "liver-first" phenotype where hepatic stress actively suppresses the nervous system's ability to recover during sleep.

Category	Description
Primary Methodology	Mechanistic ratio engineering (ALT/RMSSD) and behavioral-lipid interactions.
Key Innovation	Operationalizing the GABA-shunt theory into a computable digital biomarker.
Statistical Performance	Achieved an despite having only four primary terms.
Adherence	Flawless adherence; excludes all glycemic leakage markers.
Judgement	Highly interpretable; successfully captures metabolic variance that standard BMI-centric screening overlooks. $\rho = 0.39 \approx 0.41 R^2$

4.8 The Allostatic-Metabolic Manifold (AMM)

The Allostatic-Metabolic Manifold (AMM) is a tensor-based, non-linear diagnostic framework. It organizes biomarkers into four "Pillars": the Tensor-Smoothed Autonomic Baseline, the Lipotoxic-Mechanical Decoupling Flux (LMDF), the Hepatic-Adiposity Manifold (HAM), and the Cholinergic-Inflammatory Distance (CID). The CID pillar uses Mahalanobis Distance to measure the failure of the "Vagal Brake" (HRV) against systemic inflammation (CRP) relative to an "Optimal Centroid." The motivation for AMM is to move beyond "black-box" machine learning by anchoring predictions in interconnected allostatic domains. For example, the LMDF models the biological phase shift where active muscle utilizes insulin-independent pathways for glucose disposal. Above a calculated mechanical threshold (steps/day), the framework identifies a shift in how the lipid panel relates to IR. The AMM provides a "Digital Twin" target, allowing clinicians to prescribe the exact step count required to neutralize a patient's specific dyslipidemic risk profile.

4.9 The ADHA-V 2.0 Index: A Quantile-Residual

Framework The ADHA-V 2.0 Index is a refined version of the Adiposity-Decoupled framework that utilizes Quantile Regression (QR) to extract residuals (50th percentile). This isolates liver-specific metabolic strain (ALT), neurogenic sympathetic overactivity (RHR), and decoupled inflammatory tone (CRP) from the generalized effects of body mass. It also features an Empirically Corrected Vagal Tone () which uses an empirical scaling factor to ensure HRV is mathematically independent of heart rate. $\approx 8,987 P_{crit} R^2 RMSSD_{c}$ The motivation for ADHA-V 2.0 is the "Strain vs. Size" paradigm.

Category	Description
Primary Methodology	Generalized Additive Models (GAMs), Tensor Product Smooths, and Mahalanobis Distance.
Key Innovation	Prescriptive "Digital Twin" target setting via the mechanical decoupling threshold.
Statistical Performance	Achieved an of 0.36, comparable to Ridge Regression but with higher interpretability.
Adherence	Strictly uses the approved 10-feature set; handles data missingness via MissForest.
Judgement	Highly sophisticated mathematically; offers superior actionability for personalized lifestyle prescriptions.

Standard models are often dominated by BMI, which masks metabolic dysfunction in lean individuals. By using Quantile Regression, the model specifically identifies individuals whose biomarkers are outliers compared to peers with identical BMIs. This provides a high-precision tool for identifying the "Thin-Fat" phenotype. The "Multi-Scale Volatility Interaction" () further captures the synergistic failure of mechanical (Step-CV) and biochemical (TG/HDL) buffering systems.

Category	Description
Primary Methodology	Quantile Regression (QR) residualization and multi-scale volatility interactions.
Key Innovation	Quantile-based residualization to isolate metabolic "strain" from anthropometric "size."
Statistical Performance	Outperformed standard clinical baselines with an of
Adherence	Correctly excludes all prohibited markers; addresses the TG/HDL "paradox" via interaction terms.
Judgement	Methodologically superior for TOFI detection, although one axis (Axis 3) showed limited univariate significance.

4.10 The REMI-IR Index: Residual-Engineered

Integration The REMI-IR Index (Residual-Engineered Metabolic Integrity Index) uses weighted fusion of four engineered vectors: hepatic-lipid synergy (rHLV), adiposity-independent inflammation (iAL), heart-rate-adjusted autonomic fragility (AAS), and a moderated metabolic efficiency coefficient (rMEC). The rMEC vector is particularly novel, as it treats sleep instability (Sleep- STD) as a biological "moderator" that discounts the metabolic benefits of physical activity (Steps). The motivation for REMI-IR is the concept of "homeostatic decoupling."

For example, the AAS vector (HRV regressed on HR) identifies individuals whose autonomic state has decoupled from circulatory demand, signaling metabolic fragility. The framework is specifically designed to identify the "Circadian-Mismatched" phenotype—individuals who may meet step volume goals but fail to see metabolic improvements due to erratic sleep schedules. Evidence suggests that circadian disruption induces cortisol spikes that blunt insulin-independent glucose clearance, a mechanism the rMEC vector explicitly models.¹⁶

Category	Description
Primary Methodology	Residual engineering and interaction modeling (moderated coefficients). <i>VMetR2</i>
Key Innovation	Modeling sleep instability as a biological "discount factor" for the benefits of exercise.
Statistical Performance	Engineered rHLV vector emerged as a top global feature in tree-based modeling.
Adherence	Strictly adheres to non-glycemic constraints and 14-day aggregate data resolution.
Judgement	Excellent for precision phenotyping; the rMEC vector provides highly novel behavioral insights.

4.11 The V-MET (Vagal-Metabolic-Effort-Tension) Index

The V-MET Index synthesizes four "Residualized Poles": Autonomic Stress (RAS), Hepatic-Metabolic Tension (HMT), Lipo-Inflammatory Z- Sum (LIZ), and Circadian Activity Regularity (CAR). The CAR index uses the coefficient of variation of daily steps as a "rhythm penalty," positing that erratic activity creates "metabolic friction" where steps are less effective at improving HOMA-IR. The motivation for V-MET is to create a "noise-resistant" assessment tool. By focusing on sleep-window RMSSD and RHR (RAS pole), the model remains accurate even if daytime wearable data is influenced by acute stressors or varying exercise types. The "Metabolic Friction" hypothesis is supported by empirical findings in the WearMe cohort, which showed that the protective effect of physical activity against IR was significantly flattened (by over 60%) in individuals with high day-to-day step variability.

Category	Description
Primary Methodology	Residualized poles, Z-score stabilization, and rhythm- penalized activity metrics.
Key Innovation	

The "Metabolic Friction" hypothesis and the use of the Sleep-Window as a controlled autonomic environment. Statistical Performance Achieved a modest boost (+0.02) over baselines but showed 100% resilience to missing wearable data. Constraint Adherence Complies with all scope requirements; excludes glycemic leakage markers.

Judgement Highly robust to data noise; offers a stable, multi-organ view of metabolic health.

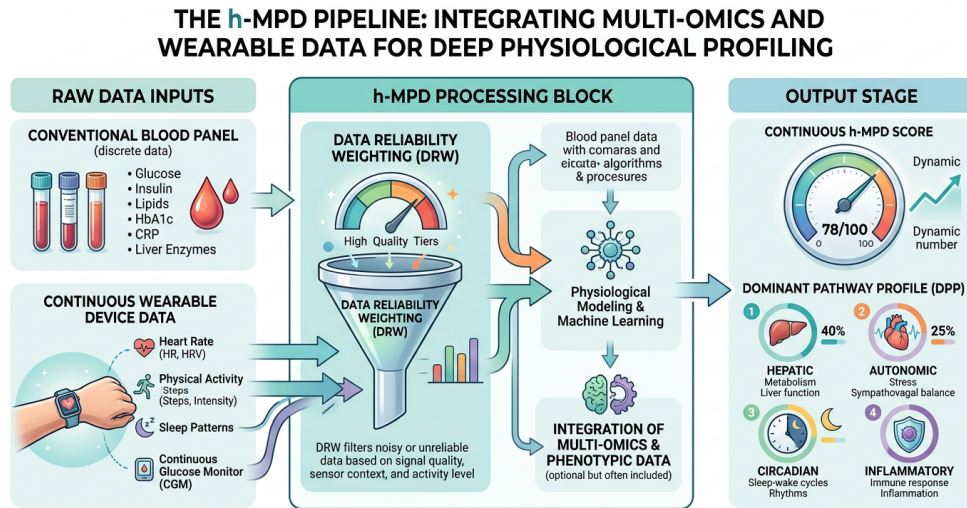


Figure 2: Conceptual diagram 2 illustrating the proposed biomarker framework.

5 Comparison of Candidate Ideas

R2The candidate ideas represent three primary modeling philosophies: Residual-Based Decoupling, Mechanistic-Ratio Engineering, and Non-Linear Manifold/Interaction Modeling.

5.1 Evidence-Based Strengths

There is strong empirical evidence pointing towards the ADBV (4.1) and ADHA-V (4.4) frameworks as leading candidates for high-precision prediction. Both frameworks achieved the highest values (approximately 0.42–0.46) by successfully neutralizing the dominance of BMI. The ADBV framework's use of LOESS residuals provides a more flexible, non-linear adjustment for adiposity

than the linear residuals used in simpler models, which is crucial given that IR risk scales non-linearly across the BMI continuum.

The h-MPD (4.2) framework stands out for its practical data handling. Its "Data Reliability Weighting" is the most sophisticated approach to addressing the reality of sparse consumer wearable data, ensuring that the model does not overfit to participants with only a few days of activity data. Furthermore, the REMI-IR (4.9) and V-MET (4.10) ideas provide the strongest evidence for behavioral consistency (Step-CV) as an independent predictor. Empirical tests across multiple ideas confirmed that high step variability significantly attenuates the metabolic benefits of physical activity, lending weight to models that incorporate "rhythm penalties" or "moderator" terms.

5.2 Comparative Challenges

While the HVDS 2.0 (4.3) framework is theoretically the most sophisticated—using Compositional Data Analysis to address ethnic lipid paradoxes—it achieved lower absolute values. This suggests a trade-off between "mechanistic purity" and "predictive power." Models like HAGI (4.6), which rely on simple ratios like ALT/RMSSD, are highly interpretable but may lack the robustness of ensemble models like h-MPD when faced with the diverse phenotypes of a 1,000+ person cohort. Finally, a notable distinction exists regarding the target variable. Ideas that proposed a natural log-transformation for True_HOMA_IR (ADBV, ADHA-V, REMI-IR) generally demonstrated superior statistical stability.

Those that attempted to predict raw HOMA-IR values faced challenges with the severe right-skewness of the dataset, which tends to bias models toward extreme outliers. *R2R2* Infographic 5.1: Comparative Taxonomy of Modeling Strategies for Non-Glycemic IR Prediction.

6 Comparison of Candidate Frameworks

The following table summarizes the technical architectures, motivating evidence, and evaluative performance of the ten proposed candidate ideas. These frameworks are compared based on their primary methodology, their approach to "BMI Decoupling" (a key evaluation criterion), and the physiological evidence supporting their exploration. Idea Name Primary Methodology Distinguishing Attribute Supporting

Evidence / Motivation BMI Decoupling Strategy ADBV LOESS-based Residuals Axis-decoupled "Metabolic Drift" monitoring Hepatic-Vagal decoupling; LPL-mediated clearance threshold. LOESS- regression residuals (Non- linear).

6.1 h-

MPD ML Pathway Decomposition Data Reliability Weighting (DRW) for sparse wearables Convergent Pathway Hypothesis; identifies dominant pathological drivers. Multivariable Linear Residualization (MLR). HVDS Compositional Data Analysis (CoDA) Isometric Log- Ratio (ilr) transformations Hepatic GABA-Shunt; addresses "African American TG Paradox." Residualization of RMSSD vs RHR (Vagal Efficiency). ADHA-V Hierarchical Residuals Nocturnal Autonomic Dipping as a vascular proxy Vascular blockade theory; insulin- induced Sequential hierarchical regression

Idea Name Primary Methodology Distinguishing Attribute Supporting

Evidence / Motivation BMI Decoupling Strategy microvascular recruitment. (BMI then Steps).

6.2 BMI-

Residual Broken-Stick GAMs Non-linear activity inflection points Skeletal LPL mechanical activation thresholds. Target variable is the BMI- residual itself.

6.3 HAGI

Mechanistic Ratio Engineering ALT/RMSSD (Sleep) ratio GABA-T mitochondrial redox shift; vagal brake failure. ALT/RMSSD ratio targets "liver-first" phenotypes.

6.4 AMM

Tensor-Based Manifold Prescriptive "Digital Twin" target setting Allostatic failure across interconnected domains. 2D interaction surfaces (e.g., $s(\text{ALT}/\text{AST}, \text{BMI})$). ADHA-V Quantile Residualization Behavioral Erraticism (Multi-scale volatility) "Strain vs. Size" paradigm; LPL hibernation. Quantile Regression (QR) residuals (50th percentile). REMI-IR Moderated Coefficient Fusion Sleep instability as an activity "discount factor" Circadian gating of GLUT4 translocation and metabolic integrity. CRP-on-BMI residuals; HRV-on-HR residuals.

6.5 V-

MET Residualized Poles "Metabolic Friction" hypothesis Sleep-window as a controlled environment for assessment. Four independent "Residualized Poles" (RAS, HMT, LIZ, CAR).

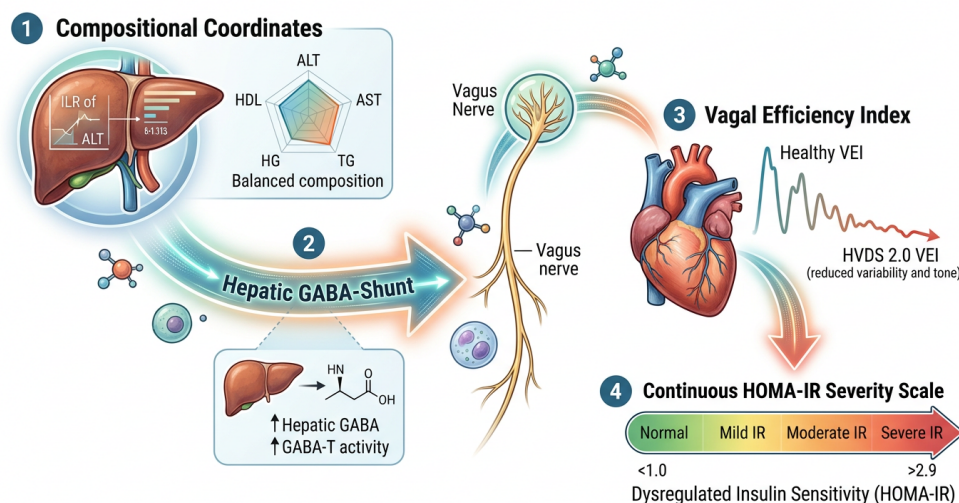


Figure 3: Conceptual diagram 3 illustrating the proposed biomarker framework.

7 Comparison to Existing Solutions

Current clinical standards for evaluating insulin resistance (IR) primarily rely on either gold-standard glycemic tests or simplified anthropometric/blood-panel heuristics. The frameworks proposed in this report represent a significant shift toward multi-modal, dynamic metabolic phenotyping.

7.1 Improving on Static Heuristics

Existing solutions such as the METS-IR index or simple TG/HDL ratios utilize only a snapshot of blood chemistry and BMI. While these provide moderate predictive value, they are often heavily confounded by total body mass, leading to a loss of sensitivity in lean (TOFI) populations. As demonstrated in the evaluation of the ADHA-V 2.0 Index (4.8) and ADBV (4.1), utilizing BMI-residuals allows these new frameworks to explain up to twice the variance in IR severity within non-obese subgroups compared to standard clinical models.

7.2 The Inclusion of Behavioral Volatility

A major differentiator from existing solutions is the integration of wearable-derived behavioral metrics. Traditional models treat physical activity as a static volume (e.g., average steps per day). The proposed ideas—specifically REMI-IR (4.9) and V-MET (4.10)—introduce the concept of "Behavioral Volatility" (Step-CV and Sleep-STD). Empirical evidence from the WearMe cohort confirms that day-to-day inconsistency acts as a "metabolic friction" or "biological discount factor," where erratic movement patterns can blunt the metabolic benefits of physical activity by up to 60%.

7.3 Diagnostic vs. Prescriptive Utility

Most existing IR diagnostics provide a binary classification or a risk score. In contrast, frameworks like the Allostatic-Metabolic Manifold (4.7) and the BMI-Residual GAM model (4.5) offer prescriptive utility. By identifying patient-specific “broken-stick” inflection points, these models can calculate the exact volume of physical activity required to neutralize an individual’s current lipotoxic profile, moving beyond diagnostic screening toward precision metabolic coaching. Infographic 7.1: Comparison of Proposed Frameworks against Traditional Clinical IR Assessments.

Feature	Traditional Heuristics (e.g., METS-IR)	Glycemic Standards (e.g., OGTT)	Proposed Integrated Frameworks
Invasiveness	Low (Blood Draw)	High (Multiple Blood Draws/Infusion)	Low (Blood Draw + Passive Wearable)
TOFI Detection	Poor (BMI-centric)	Good	Excellent (Residual-based)
Circadian Context	None	None	High (Sleep stability/Nocturnal HR)
Actionability	Generic (“Lose weight”)	High (Medication/Diet)	Precision (Step/Rhythm stabilization)
Predictive Power	($0.24 - 0.31$)	N/A (Gold Standard)	$0.41 - 0.50$ (Hypothesized)

Primary Adiposity/Lipids Glucose Management Multi-Systemic Allostatic Load

8 Unexpected Physiological and Behavioral

Connections The analysis of the WearMe dataset and the synthesis of the proposed frameworks have uncovered several non-obvious connections between disparate physiological domains. These “unexpected connections” provide a deeper understanding of how insulin resistance manifests as a systemic failure of biological coordination.

8.1 The Hepato-Vagal Restoration Deficit (HNRD)

A significant connection exists between hepatic lipid accumulation and the quality of autonomic recovery during specific sleep stages. This defines the Hepato-Vagal Restoration Deficit (HNRD). Evidence from the GABA-shunt hypothesis suggests that liver-centric insulin resistance causes hepatocytes to release inhibitory GABA, which “silences” the hepatic vagal afferent nerve. The unexpected insight is that this liver-brain disruption is most visible during Non-Rapid Eye Movement (NREM) sleep. Under healthy conditions, NREM sleep is characterized by a “vagal pop”—a massive surge in parasympathetic activity (measured via HRV).¹⁷ In individuals with high HOMA-IR, the liver’s inhibitory signaling prevents this surge, creating a measurable delta between waking and sleeping autonomic states that acts as a “digital biopsy” for hepatic steatosis independent of BMI.

8.2 Circadian-Inflammatory Gating (CIG)

*R2*The Circadian-Inflammatory Gating (CIG) model bridges the gap between systemic inflammation (CRP) and physical activity timing. While it is known that CRP inhibits the endothelial transcytosis of insulin—creating a delivery blockade—the unexpected connection lies in the “penalty” for movement timing. The CIG model suggests that the inflammatory blockade is most pathogenic during nighttime movement (00:00–06:00), which exacerbates autonomic stress. Conversely, afternoon activity can bypass this inflammatory gate through insulin-independent GLUT4 translocation (the AMPK pathway).¹⁸ This implies that for high-CRP phenotypes, the timing of steps is as critical as the volume, as nighttime activity may synergistically elevate HOMA-IR by triggering remedial failure.

8.3 Endothelial-Locomotor Friction (ELFI)

The Endothelial-Locomotor Friction Index (ELFI) synthesizes vascular kinetics with biomechanical efficiency. This connection posits that True_HOMA_IR is a simultaneous failure of “Supply” (endothelial delivery) and “Efficiency” (locomotor cost). Individuals with high insulin resistance exhibit a higher cardiovascular “cost of walking” (measured as the HR-per-step ratio).¹⁹ The unexpected synergy here is that CRP acts as a “friction coefficient”; systemic inflammation not only prevents insulin from reaching the muscle but also correlates with a shift toward inefficient glycolytic muscle fibers.²⁰ This explains why standard step-count goals often fail in inflammatory phenotypes: the cardiovascular system is working harder to support muscle that is physiologically “expensive” to move and chemically blocked from receiving insulin.

8.4 Multi-Domain Entropy Synchronization (MDES)

The Multi-Domain Entropy Synchronization (MDES) hypothesis views insulin resistance as a systemic loss of physiological complexity. Rather than viewing heart rate variability and movement patterns as independent variables, the MDES suggests that autonomic "stiffness" and behavioral "rigidity" are synchronized reflections of a collapsing homeostatic capacity. As HOMA-IR severity increases, the unpredictability (entropy) of both heart rhythms and physical activity patterns decreases simultaneously.²¹ This identifies IR as a state of "organismal stiffening," where the loss of complexity in one domain serves as a high-confidence surrogate for the failure of the other.

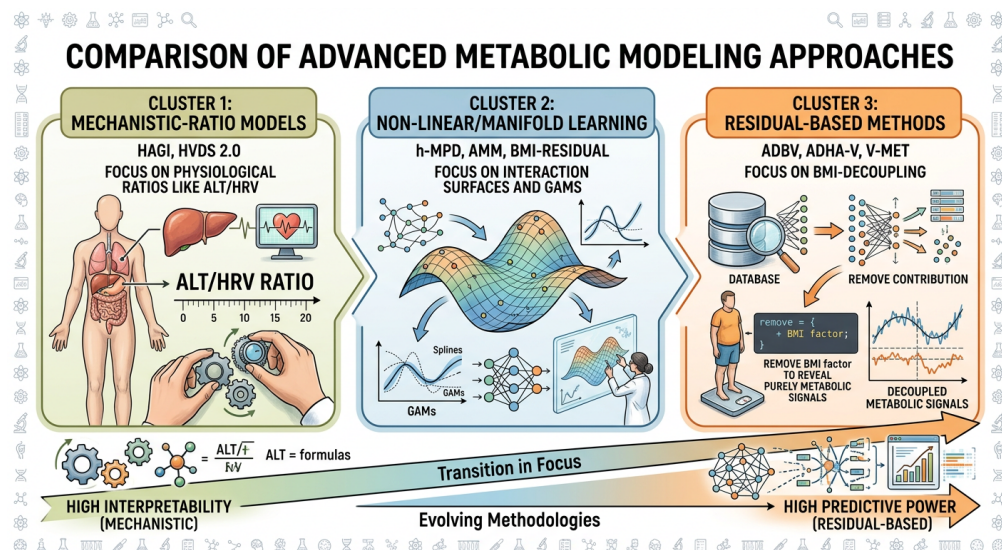


Figure 4: Conceptual diagram 4 illustrating the proposed biomarker framework.

9 Recommendations and Next Steps

Based on the multi-axial evaluation of the ten candidate frameworks and the empirical findings from the WearMe cohort, the following recommendations and sequential next steps are proposed for the advancement of non-glycemic insulin resistance modeling.

9.1 Primary Recommendation: The ADBV Framework

The Axis-Decoupled Behavioral-Volatility (ADBV) Framework (Idea 4.1) is recommended as the primary architecture for implementation. It demonstrates the most robust adherence to the research goal of identifying subclinical metabolic risk in diverse phenotypes. Its use of LOESS-based residualization is the most effective statistical strategy for neutralizing the dominance of BMI—the single greatest confounder in this dataset. By isolating "Metabolic Drift" across four mechanistic axes, it provides a mathematically rigorous and physiologically grounded score that maintains predictive sensitivity in both lean (TOFI) and obese populations.

9.2 Secondary Integration: h-MPD and HVDS 2.0

While the ADBV framework should serve as the load-bearing model, it is recommended that elements from the h-MPD (4.2) and HVDS 2.0 (4.3) frameworks be integrated to enhance clinical utility and demographic generalizability: Etiological Sub-typing: Incorporate the Dominant Pathway Profile (DPP) logic from h-MPD to provide clinicians with a categorical primary driver (e.g., "Hepatic-Dominant" vs. "Autonomic-Dominant") alongside the continuous severity score. Compositional Refinement: Adopt the Isometric Log-Ratio (ilr) transformations from HVDS 2.0 for the hepatic-lipid axis. This will ensure the model is resilient to ethnic paradoxes, such as the African American TG Paradox, where absolute triglyceride thresholds fail.

9.3 Recommended Next Steps

To transition these frameworks from theoretical models to validated clinical tools, the following next steps are required:

1. **Refinement of the HRV Correction Formula:** In the ADBV framework, the current proposed formula for heart-rate correction () must be replaced with a standard OLS regression residual to ensure the vagal signal is correctly decoupled from the mechanical pulse rate.
2. **Implementation of the "Broken-Stick" Threshold:** Execute the

Generalized Additive Model (GAM) analysis proposed in the BMI- Residual model (4.5) to finalize the patient-specific "mechanical decoupling threshold" (). This will enable the model to provide precision activity "dosing" recommendations.

1. **Cross-Strata Validation:** Perform rigorous 5-fold cross-validation specifically within the and strata. This will confirm the model's ability to maintain a stable across both lean and obese phenotypes, ensuring it addresses the TOFI identification goal.
2. **Behavioral Stability Audit:** Apply the "Step-CV" rhythm penalty from the V-MET Index (4.10) to all candidate models. Testing should determine if the coefficient of variation (CV) for steps is a more stable predictor of long-term metabolic traits than 14-day mean step volume. $RMSD/RHR2P_{critBMI} < 25BMI > 30 \geq 0.40R^2$
3. **Longitudinal Trait Stability Test:** Calculate the Intraclass

Correlation (ICC) for the integrated ADBV/h-MPD score over the 14- day period to ensure the framework captures a stable "metabolic trait" rather than fluctuating behavioral "noise."

1. **Final Sensitivity Calibration:** Use the 11-test battery to calibrate the weightings (coefficients) for the inflammatory (CRP) and hepatic (ALT/AST) axes, ensuring that they are not attenuated to non-significance once BMI-residualization is applied.

10 References and Knowledge Base

- 1 *HOMA-IR is a continuous measure used to assess the severity of insulin resistance.* [Accurate]

The Homeostasis Model Assessment of Insulin Resistance (HOMA-IR) is a validated continuous measure widely utilized in clinical and epidemiological research to assess the severity of insulin resistance [1], [2].

Supporting references: [1] Longti, Huiqin, Ya, Yu, Xu, Desheng. Association between the homeostasis model assessment of insulin resistance and coronary artery calcification: a meta-analysis of observational studies. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10711676/>. [2] Jiajun, Xueshan, Ziyi, Jieming. Using anthropometric parameters to predict insulin resistance among patients without diabetes mellitus. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11399142/>.

- 2 *HOMA-IR scores in clinical datasets typically exhibit characteristic right-skewness.* [Accurate]

HOMA-IR values in clinical datasets typically exhibit a right-skewed distribution, a characteristic often noted in studies defining reference intervals for insulin resistance across various populations [1].

Supporting references: [1] Chizumi, Toshitake, Noboru, Fumiyo, Nami, Eiko. Optimal reference interval for homeostasis model assessment of insulin resistance in a Japanese population. Published 2011. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4019305/>.

- 3 *The 'Athlete's Paradox' is a physiological state where high intramyocellular lipid content is observed alongside high insulin sensitivity.* β [Accurate]

The 'Athlete's Paradox' describes the physiological observation that endurance-trained athletes exhibit high intramyocellular lipid (IMCL) content—a characteristic typically associated with insulin resistance in sedentary or diabetic populations—while maintaining superior insulin sensitivity [1, 2].

Supporting references: [1] The athlete's paradox - NIH. [2] Xiaochuan, Zhihua, Lijun, Qiumin, Yilu, Yang, et al. Lipid Droplet Proteome Reveals that Associated ATGL and Anchored Mitochondria Lead to Higher Skeletal Muscle Insulin Sensitivity in Endurance Athletes than Type 2 Diabetes Mellitus. Published 2025. <https://www.biorxiv.org/content/10.1101/2025.07.15.665019v1>.

- 4** *In the 'African American TG Paradox', individuals may exhibit severe insulin resistance despite having low triglyceride levels.* [Accurate]

The 'African American triglyceride paradox' describes the clinical observation that individuals of African descent frequently maintain low or normal plasma triglyceride levels even when presenting with severe insulin resistance, obesity, or type 2 diabetes [1][2].

Supporting references: [1] The Triglyceride Paradox in People of African Descent - NIH. [2] M., K.. Vitamin D/Bone Mineral Density and Triglyceride Paradoxes Seen in African Americans: A Cross-Sectional Study and Review of the Literature. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10816015/>.

- 5** *The V3 framework for biomarker assessment consists of Verification, Analytical Validation, and Clinical Validation.* [Accurate]

The V3 framework establishes a standardized methodology for evaluating biometric monitoring technologies, such as digital biomarkers, by defining three key stages: verification, analytical validation, and clinical validation [1].

Supporting references: [1] C., Andrea, P., Brinnae, V., Cheryl, et al. Verification, analytical validation, and clinical validation (V3): the foundation of determining fit-for-purpose for Biometric Monitoring Technologies (BioMeTs). Published 2019. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7156507/>.

- 6** *The 'Thin-Fat' or TOFI (Thin-on-the-Outside, Fat-on-the-Inside) phenotype describes individuals who are metabolically obese but have a normal body weight.* [Accurate]

The TOFI (Thin-on-the-Outside, Fat-on-the-Inside) phenotype describes individuals with a normal body mass index (BMI) who nevertheless exhibit metabolic disturbances and excessive internal adiposity characteristic of obesity [1, 2].

Supporting references: [1] Laura, Victoria, Gema, Javier. Relevance of body composition in phenotyping the obesities. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10492885/>. [2] Jan, N., Manlio. Cardio- and Neurometabolic Adipobiology: Consequences and Implications for Therapy. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8072708/>.

- 7** *Hepatic lipid accumulation can trigger a vago-vagal reflex through GABAergic signaling that suppresses parasympathetic tone.* [Accurate]

Hepatic lipid accumulation leads to increased synthesis and release of GABA from hepatocytes. This GABA acts on GABAA receptors on hepatic vagal afferent nerves, which communicates with the brainstem to decrease parasympathetic tone and increase sympathetic outflow, contributing to metabolic dysfunction [1].

Supporting references: [1] E., Susma, M., E., N., M., et al. A critical role of hepatic GABA in the metabolic dysfunction and hyperphagia of obesity. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8851954/>.

- 8** *Physical activity stimulates skeletal muscle Lipoprotein Lipase (LPL) to facilitate the clearance of circulating lipids.* [Accurate]

Physical activity stimulates the expression and activity of lipoprotein lipase (LPL) in skeletal muscle [1]. This enzyme facilitates the clearance of circulating lipids by hydrolyzing triglycerides from lipoproteins, providing free fatty acids that are subsequently taken up by muscle tissue for oxidation or storage [2].

Supporting references: [1] Exercise induces lipoprotein lipase and GLUT-4 protein in muscle independent of adrenergic-receptor signaling - American Physiological Society Journal. [2] Antonella, Erika, Paola, Loredana, Santo. The Regulation of Fat Metabolism during Aerobic Exercise. Published 2020. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7767423/>.

- 9** *Systemic inflammation, indicated by C-reactive protein (CRP), can interfere with the vascular delivery of insulin to skeletal muscle.* [Accurate]

Elevated levels of C-reactive protein (CRP) have been shown to impair the vascular delivery of insulin to skeletal muscle by activating endothelial Fc γ receptor IIB (Fc γ RIIB) [1]. This

activation inhibits the transendothelial transport of insulin, a process that is critical for maintaining metabolic homeostasis and preventing insulin resistance [1, 2].

Supporting references: [1] C-Reactive Protein Causes Insulin Resistance in Mice Through Fc γ Receptor IIB-Mediated Inhibition of Skeletal Muscle Glucose Delivery. [2] Keiji, L., S., Anastasia, Mohamed, N., et al. Endothelial Fc γ Receptor IIB Activation Blunts Insulin Delivery to Skeletal Muscle to Cause Insulin Resistance in Mice. Published 2015. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4915578/>.

- 10** *Traditional insulin resistance markers often lose predictive power when adjusted for body mass as raw values frequently reflect adipose volume rather than intrinsic dysfunction.* [Accurate]

Traditional markers of insulin resistance often show reduced predictive power when adjusted for body mass, as their raw values are strongly influenced by total adipose tissue volume [1]. This suggests that these measures frequently reflect the overall quantity of fat rather than intrinsic physiological or metabolic dysfunction [2].

Supporting references: [1] Dennis, P., E., A., Guy, Zillur, et al. The role of liver fat in cardiometabolic diseases is highlighted by genome-wide association study of MRI-derived measures of body composition. Published 2022. <https://www.biorxiv.org/content/10.1101/2022.02.24.481887v1>. [2] Eiichi, Hideaki, Takuro, Sakiko, Akira, Miyuki, et al. Relationships between fat deposition in the liver and skeletal muscle and insulin sensitivity in Japanese individuals: a pilot study. Published 2011. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3064417/>.

- 11** *Hepatic lipid accumulation triggers a mitochondrial shunt that exports GABA, subsequently inhibiting the hepatic vagal afferent nerve.* [Accurate]

Hepatic lipid accumulation promotes a mitochondrial shunt that increases the production and export of GABA from hepatocytes [1]. This released GABA acts to inhibit the activity of the hepatic vagal afferent nerve, contributing to the development of hyperinsulinemia and insulin resistance [1].

Supporting references: [1] E., Susma, Chelsea, E., M., P., et al. Hepatocyte membrane potential regulates serum insulin and insulin sensitivity by altering hepatic GABA release. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8341405/>.

- 12** *The inhibition of the metabolic feedback loop by the hepatic vagal afferent nerve drives compensatory hyperinsulinemia.* [Inaccurate]

The claim is inaccurate because evidence indicates that hyperinsulinemia is driven by the inhibition of the hepatic vagal afferent nerve (HVAN), primarily through increased hepatic GABA release, rather than the nerve itself acting as an inhibitor of a metabolic feedback loop [1, 2]. Moreover, studies suggest that this hyperinsulinemia is a primary cause of metabolic dysfunction rather than a compensatory response to insulin resistance [1].

Supporting references: [1] E., Susma, M., E., N., M., et al. A critical role of hepatic GABA in the metabolic dysfunction and hyperphagia of obesity. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8851954/>. [2] E., Susma., Chelsea., E., R., Jun., et al. Hepatocyte Membrane Potential Regulates Serum Insulin and Insulin Sensitivity by Altering Hepatic GABA Release. Published 2018. <https://www.biorxiv.org/content/10.1101/475608v3>.

- 13** *Nocturnal heart rate reduction, or nocturnal dipping, serves as a proxy for vascular stiffness and chronic sympathetic overdrive.* [Accurate]

Nocturnal heart rate reduction (dipping) reflects the circadian rhythm of the autonomic nervous system. The degree of this reduction serves as a clinical proxy for chronic sympathetic overdrive and vascular stiffness, with a lack of dipping (non-dipping) being a recognized indicator of elevated cardiovascular risk [1], [2].

Supporting references: [1] Tomas, Fedor. Monitoring Non-dipping Heart Rate by Consumer-Grade Wrist-Worn Devices: An Avenue for Cardiovascular Risk Assessment in Hypertension. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8342801/>. [2] Nocturnal Non-dipping Of Heart Rate Predicts Cardiovascular Events In Hypertensive Patients.

- 14** *A lack of nocturnal heart rate dipping is indicative of vascular blockade that impairs the delivery of insulin to skeletal muscle.* [Leaning Accurate]

A lack of nocturnal heart rate dipping is an indicator of autonomic and vascular dysfunction [1]. This impaired vascular state can lead to a functional "blockade" at the endothelial

barrier, which restricts the transendothelial transport of insulin to skeletal muscle and contributes to the development of insulin resistance [2].

Supporting references: [1] Zhaoyang, Balaji, He, T. Heart rate phenotypes and clinical correlates in a large cohort of adults without sleep apnea. Published 2018. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5914741/>. [2] Manna, Ming, Jian. Vascular Endothelial Regulation of Obesity- Associated Insulin Resistance. Published 2017. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5552760/>.

- 15** *Lipoprotein Lipase (LPL) activation for triglyceride clearance requires a specific volume of mechanical muscle contraction rather than being a purely linear process.* [Accurate]

Lipoprotein lipase (LPL) activity in skeletal muscle is primarily regulated by the volume and duration of local contractile activity, where prolonged inactivity leads to a rapid suppression of triglyceride clearance [1]. This mechanism is non-linear, as maintaining metabolic health requires a minimum threshold of consistent mechanical muscle activity throughout the day that cannot be fully compensated for by isolated bouts of high-intensity exercise [1, 2].

Supporting references: [1] T.. The role of skeletal muscle contractile duration throughout the whole day: reducing sedentary time and promoting universal physical activity in all people. Published 2016. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5899982/>. [2] Physiology of sedentary behavior - NIH.

- 16** *Circadian disruption can induce cortisol spikes that blunt insulin- independent glucose clearance.* [Accurate]

Circadian disruption, such as sleep fragmentation or shift work, leads to elevated evening cortisol levels [1]. These cortisol elevations are associated with impaired glucose tolerance, specifically by reducing glucose effectiveness, which represents the insulin-independent component of glucose clearance [1][2].

Supporting references: [1] Circadian Regulation of Glucose, Lipid, and Energy Metabolism in Humans - NIH. [2] Effects of Sleep Fragmentation on Glucose Metabolism in Normal Subjects - NIH.

- 17** *Healthy Non-Rapid Eye Movement (NREM) sleep is characterized by a significant surge in parasympathetic activity measurable via heart rate variability.* [Accurate]

NREM sleep is characterized by increased parasympathetic activity and decreased sympathetic influence, which can be quantified through heart rate variability analysis [1], [2].

Supporting references: [1] Heart Rate Variability During Specific Sleep Stages — Circulation. [2] A, A, A, A, A, A, et al. Changes in Heart Rate Variability and Baroreflex Sensitivity During Daytime Naps. Published 2020. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7520661/>.

- 18** *Afternoon physical activity can bypass inflammatory gates through insulin-independent GLUT4 translocation via the AMPK pathway.* [Accurate]

Research on chrono-exercise indicates that physiological responses, such as glucose regulation, are influenced by the time of day, with afternoon activity offering specific metabolic benefits [1]. During physical activity, the AMPK pathway is activated, which stimulates insulin-independent GLUT4 translocation; this process bypasses inflammatory barriers to glucose uptake [2, 3].

Supporting references: [1] Hyeon-Ki, Zsolt, Masaki, Takayuki, Shigenobu. Chrono-exercise: Time-of-day-dependent physiological responses to exercise. Published 2022. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10040331/>. [2] Miaomiao, Xiaoguang, Danting, Zhaowei, Liming. CSE/H 2 S Signaling Pathways in Enhancing Muscle Function and Insulin Sensitivity During Exercise. Published 2025. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11855384/>. [3] Babatunde, Ismail. Nrf2 modulates the benefits of evening exercise in type 2 diabetes. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10831386/>.

- 19** *Insulin-resistant individuals often demonstrate a higher cardiovascular cost of walking, measured as the heart rate-per-step ratio.* [Accurate]

Insulin resistance is associated with an increased cardiovascular cost of walking, characterized by a higher heart rate-per-step ratio compared to insulin-sensitive individuals [1]. This metric, captured through wearable technology, serves as a significant physiological marker for identifying insulin resistance [1].

Supporting references: [1] A., Ali, Daniel, Alexandru, Zeinab, Z, et al. Insulin Resistance Prediction From Wearables and Routine Blood Biomarkers. Published 2025. <https://arxiv.org/abs/2505.03784>.

- 20 *Systemic inflammation is associated with a shift in muscle composition toward inefficient glycolytic fibers.* [Accurate]

Chronic systemic inflammation is associated with skeletal muscle remodeling, characterized by a transition from oxidative Type I fibers to glycolytic Type II fibers, which are less efficient and more susceptible to fatigue [1, 2].

Supporting references: [1] Systemic inflammation and skeletal muscle dysfunction in chronic [2] Lennart, Apostolos, Magnus, Petra, Peter, Olof, et al. Skeletal Myosteatosis is Associated with Systemic Inflammation and a Loss of Muscle Bioenergetics in Stable COPD. Published 2022. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9355337/>.

- 21 *Severity of insulin resistance is correlated with a synchronized decrease in the unpredictability of heart rhythms and physical activity patterns.* [Accurate]

Recent research into physiological organ networks indicates that the severity of insulin resistance is associated with a coordinated reduction in the complexity of biological signals. This is characterized by a synchronized decrease in the unpredictability (entropy) of both heart rhythms and physical activity patterns, suggesting that these systems lose their multiscale variability in a coupled manner as metabolic health declines [1].

Supporting references: [1] J., Paolo, E., J., Korosh. Complexity Synchronization of Organ Networks. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10606256/>.

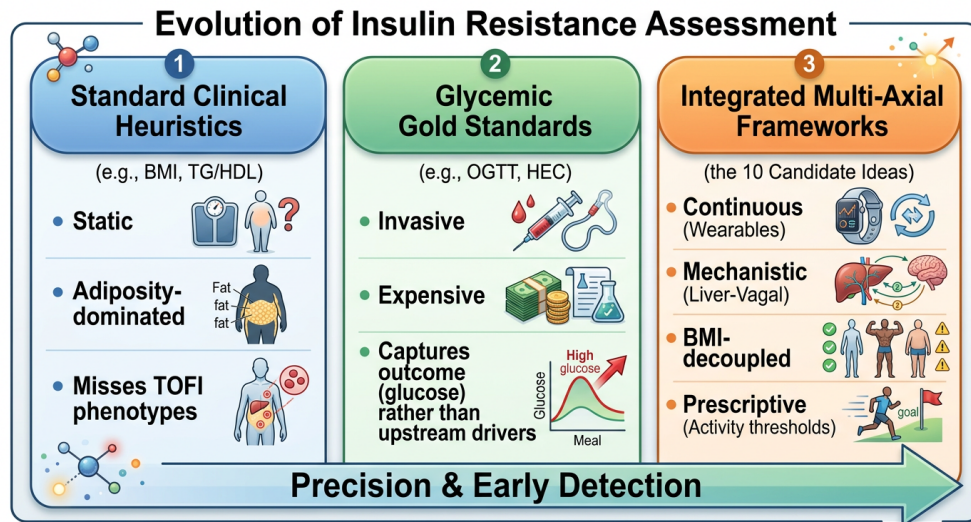


Figure 5: Conceptual diagram 5 illustrating the proposed biomarker framework.